Coupled Hyperbolic Multi-Sphere Deep SVDD

CJ
Applied CS
BTU
Germany
CJ

***Abstract*—Deep SVDD is a strong one-class objective but is commonly used in one-vs-rest form, requiring one model per class and lacking a shared multi-class geometry. We present a coupled hyperbolic multi-sphere extension that learns class-wise hyperspheres jointly in a single shared representation. Our method places class embeddings in a Poincaré ball and optimizes four coupled terms: reconstruction, class-conditional Deep SVDD compactness, inter-class exclusion, and pairwise sphere-overlap separation. This yields a single model with class-aware inlier regions and a natural global anomaly score. We evaluate on MNIST and CIFAR-10 with the same training recipe (two-stage AE then coupled SVDD).**

***Index Terms*—Anomaly detection, deep SVDD, hyperbolic embedding, open-set recognition, Poincaré ball, multi-class learning.**

I. Introduction

Ruff et al. cast unsupervised anomaly detection as learning a minimum-volume hypersphere in latent space (Deep SVDD) [1]. The common setup trains one model per normal class (one-vs-rest). For multi-class open-set use, that duplicates parameters and does not couple class boundaries in a single geometry.

We learn K class-conditional *hyperbolic* hyperspheres *jointly* in one network: a shared representation feeds K linear heads whose outputs are mapped into a Poincaré ball. Hyperbolic distances grow toward the ball boundary, which can help separate fine-grained structure inside compact regions. Beyond per-class SVDD compactness, By adding: (i) **inter-class exclusion**, penalizing points of class y that fall too close to another class k's ball; and (ii) **sphere overlap penalty**, discouraging geometric overlap between class spheres in hyperbolic space.

II. Related Work

A. Deep SVDD

Ruff et al. optimize one-class compactness in latent space with soft-boundary or one-class objectives, using a fixed center and no bias terms to mitigate trivial solutions [1].

B. Hybrid autoencoder and SVDD

Combining reconstruction with SVDD-like compactness is a common recipe to stabilize representations and avoid collapse.

C. Positioning

We retain the SVDD principle but formulate a *joint multi-class* objective in one model with explicit class coupling and hyperbolic geometry.

III. Method

A. Overview

Training is **two-stage**, consistent with hybrid AE + SVDD: (1) pretrain the autoencoder on reconstruction only; (2) run coupled optimization with SVDD, exclusion, and overlap terms while refreshing radii on a schedule (Sec. III-H). The same loss structure is used for CHMS-SVDD on MNIST and CIFAR-10; only the convolutional AE backbone changes across datasets.

B. Architecture

Given input x, a shared autoencoder outputs representation and reconstruction:

$$(\mathbf{r}, \hat{x}) = f_\theta(x), \quad \mathbf{r} \in \mathbb{R}^{d_r}. \tag{1}$$

For K classes, class k applies a linear map (no bias) and maps into the Poincaré ball $\mathbb{B}_c^d$ via the exponential map at the origin:

$$\mathbf{u}_k = W_k \mathbf{r}, \qquad \mathbf{z}_k = \exp_0^c(\mathbf{u}_k), \quad k \in \{1, \dots, K\}. \tag{2}$$

All classes share encoder/decoder parameters θ; the W_k induce class-conditional embeddings in hyperbolic space.

Class parameters are center–radius pairs $\{(\mathbf{c}_k, R_k)\}$. Here $R_k > 0$ is a geodesic *radius*; objectives compare R_k^2 to squared hyperbolic distance $d_{\mathbb{H}}^2$. After Stage 1, set $\mathbf{c}_k$ to the class-conditional mean of embedded training points, projected into the ball; keep $\mathbf{c}_k$ *fixed* during Stage 2 [1]. Update R_k by the $(1-\nu)$ quantile rule (not by backprop), as in Sec. III-H.

C. Encoder–decoder backbones

The *hyperbolic multi-sphere* wrapper is the same for both datasets: it holds a backbone $f_\theta = (\mathrm{enc}_\theta, \mathrm{dec}_\theta)$ plus K linear heads $W_k \colon \mathbb{R}^{d_r} \to \mathbb{R}^{d_z}$ (bias-free) and curvature c. **MNIST** (`MNIST_LeNet_SVDDIAE`): input $1{\times}28{\times}28$; encoder is two $5{\times}5$ conv layers (filters $8{\to}4$) with batch norm (no affine), LeakyReLU, $2{\times}2$ max-pooling, then a fully connected layer to $\mathbf{r} \in \mathbb{R}^{d_r}$ with default $d_r{=}32$ (required divisible by 16 for the decoder reshape); decoder mirrors via upsampling, transposed convolutions, and a sigmoid output matching scaled inputs. **CIFAR-10** (`CifarConvAE`): input $3{\times}32{\times}32$; encoder is three $5{\times}5$ conv stacks ($32{\to}64{\to}128$ channels) with LeakyReLU and $2{\times}2$ pooling, then fc to $\mathbf{r} \in \mathbb{R}^{d_r}$ (default $d_r{=}128$); decoder maps $\mathbf{r}$ back to $128{\times}4{\times}4$ and decodes with three $4{\times}4$ transposed-conv stages to $3{\times}32{\times}32$. Default hyperbolic embedding dimension is $d_z{=}16$ on MNIST and $d_z{=}32$ on CIFAR.

D. Hyperbolic distance

For $\mathbf{x}, \mathbf{y} \in \mathbb{B}_c^d$,

$$d_{\mathbb{H}}(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{c}} \operatorname{arcosh}\left(1 + \frac{2c\|\mathbf{x}-\mathbf{y}\|^2}{(1-c\|\mathbf{x}\|^2)(1-c\|\mathbf{y}\|^2)}\right). \tag{3}$$

E. Objective

For labeled sample (x_i, y_i), define class-conditional embeddings $\mathbf{z}_{i,k}$ and hyperbolic squared distances to fixed centers:

$$\mathbf{z}_{i,k} = \exp_0^c(W_k f_\theta(x_i)), \quad d_{i,k}^2 = d_{\mathbb{H}}(\mathbf{z}_{i,k}, \mathbf{c}_k)^2. \tag{4}$$

The SVDD term uses only $k = y_i$ with true hyperbolic d_{i,y_i}^2. The exclusion term sums over $k \neq y_i$, comparing each $\mathbf{z}_{i,k}$ to $\mathbf{c}_k$ (see code note below for MNIST vs. CIFAR distance).

1) Reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_i \|x_i - \hat{x}_i\|_2^2. \tag{5}$$

2) Class-conditional Deep SVDD (soft-boundary): This is the class-conditional analogue of Ruff et al.'s *soft-boundary* objective: each term adds the explicit $R_{y_i}^2$ hull term and a hinge on violations outside the sphere [1]. Here d_{i,y_i}^2 uses the hyperbolic distance from $\mathbf{z}_{i,y_i}$ to $\mathbf{c}_{y_i}$. The radius R_{y_i} is not updated by backprop; after warm-up it is set by the $(1-\nu)$ quantile rule in Sec. III-H (Eq. (10)), matching the Deep SVDD soft-boundary radius schedule.

$$\mathcal{L}_{\text{svdd}} = \frac{1}{N} \sum_i \left(R_{y_i}^2 + \frac{1}{\nu} \max\left(0,\ d_{i,y_i}^2 - R_{y_i}^2\right) \right). \tag{6}$$

(*One-class variant:* replace $\mathcal{L}_{\text{svdd}}$ by $\frac{1}{N}\sum_i d_{i,y_i}^2$.)

3) Inter-class exclusion: To push samples of class y_i away from other classes' balls, we penalize points that lie closer than a *length* margin $m_{\text{excl}} \geq 0$ beyond radius R_k, i.e. outside the sphere of radius $R_k + m_{\text{excl}}$ in the same metric. In squared-distance form (all terms have units of squared length):

$$\mathcal{L}_{\text{excl}} = \frac{1}{N(K-1)} \sum_i \sum_{k \neq y_i} \max\left(0,\ (R_k + m_{\text{excl}})^2 - d_{i,k}^2\right). \tag{7}$$

The released trainers implement this hinge on squared distances to non-true class centers. **Implementation detail:** in the MNIST CHMS-SVDD trainer, $d_{i,k}^2$ inside $\mathcal{L}_{\text{excl}}$ is computed as squared *Euclidean* distance $\|\mathbf{z}_{i,k} - \mathbf{c}_k\|_2^2$ in embedding coordinates (faster); in the CIFAR-10 CHMS-SVDD trainer, $d_{i,k}^2 = d_{\mathbb{H}}(\mathbf{z}_{i,k}, \mathbf{c}_k)^2$ for every k. The exposition above uses hyperbolic $d_{i,k}^2$ for a single consistent geometry; the hinge shape is unchanged.

4) Pairwise overlap penalty:

$$\mathcal{L}_{\text{ov}} = \frac{2}{K(K-1)} \sum_{a<b} \max\left(0,\ R_a + R_b + m_{\text{ov}} - d_{\mathbb{H}}(\mathbf{c}_a, \mathbf{c}_b)\right), \tag{8}$$

which discourages overlap of class spheres, providing a more robust and flexible multi-class geometry with interpretable class-wise boundaries.

TABLE I: Terms in the coupled objective (soft-boundary mode).

Symbol	Role
$\mathcal{L}_{\text{rec}}$	Mean squared error $\|\hat{x} - x\|_2^2$; anchors θ and prevents collapse.
$\max(0, \cdot)$	Hinge (ReLU) penalty; zero if constraint satisfied, linear penalty otherw
ν	Ruff-style parameter: at most a fraction ν of in-class points may violate also sets the $(1-\nu)$ quantile for R_k [1].
$R_{y_i}^2 + \frac{1}{\nu}(\cdot)$	Soft-boundary SVDD: fixed "volume" term R^2 plus scaled violations ou
$\lambda_{\text{svdd}}, \lambda_{\text{excl}}, \lambda_{\text{ov}}$	Trade off compactness *vs.* cross-class separation *vs.* non-overlapping cen
m_{excl}	Extra geodesic slack beyond R_k for *foreign* class balls (exclusion).
m_{ov}	Minimum gap between sums of radii and center–center distance (anti-ov
$\lambda_w \|\theta\|_2^2$	Weight decay on trainable weights (not on R_k or $\mathbf{c}_k$).

5) Total loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{svdd}}\mathcal{L}_{\text{svdd}} + \lambda_{\text{excl}}\mathcal{L}_{\text{excl}} + \lambda_{\text{ov}}\mathcal{L}_{\text{ov}} + \lambda_w\|\theta\|_2^2. \tag{9}$$

F. Optimization and hyperparameters

a) Solver: We use **Adam** [2] twice: Stage 1 runs Adam on all trainable weights but the loss depends only on the AE forward path $(\mathbf{r}, \hat{x}) = f_\theta(x)$, so gradients update encoder/decoder; bias-free heads W_k are in the optimizer state but get *zero* gradient until Stage 2. Stage 2 runs Adam with learning rate 10^{-4} (default), weight decay λ_w=10^{-6}, on θ and $\{W_k\}$. Radii R_k are *not* gradient variables: after warm-up they are overwritten each epoch via (10).

b) Schedules (defaults in code): MNIST: AE pretrain 10 epochs at lr=10^{-3}, batch size 256; coupled phase 25 epochs at lr=10^{-4}; radius quantiles apply after 5 warm-up epochs. CIFAR-10: AE 20 epochs at 10^{-3}; coupled phase 40 epochs at 10^{-4}; warm-up 8 epochs; batch 256. Evaluation on a hold-out split can be logged every 5 epochs (implementation detail).

c) Loss weights and margins (defaults): λ_{svdd}=5×10^{-5}, λ_{excl}=10^{-2}, λ_{ov}=10^{-2}, curvature c=1, ν=0.1, margins m_{excl}=0.1, m_{ov}=0.05. These trade reconstruction stability against SVDD compaction and geometric separation; they should be tuned per dataset.

G. Interpretation of objective terms

Table I summarizes notation; informal statements we adopt are listed in Sec. III-I.

H. Training protocol

Stage 1 (AE pretrain): optimize $\mathcal{L}_{\text{rec}}$ only.

Center initialization: for each class, set $\mathbf{c}_k$ to the mean of projected embeddings and project onto the ball.

Stage 2 (joint): optimize $\mathcal{L}$. For soft-boundary training, after warm-up, each *radius* R_k is set from the $(1-\nu)$-quantile of *unsquared* hyperbolic distances for class k, so that R_k^2 equals the $(1-\nu)$-quantile of squared distances (consistent with scores $d^2 - R_k^2$):

$$q_k = \text{Quantile}_{1-\nu}\left(\{\, d_{i,k}^2 : y_i = k \,\}\right), \qquad R_k \leftarrow \sqrt{q_k}. \tag{10}$$

I. Methodological claims (Currently informal)

We do *not* prove new generalization bounds; the following are explicit design claims that motivate the formulation.

1) **Single-network multiclass geometry.** One shared θ and hyperbolic heads yield a *common* latent space in which all classes have class-conditional balls; this removes one-model-per-class duplication of one-vs-rest Deep SVDD while keeping per-class scores.
2) **Coupling beyond SVDD.** Exclusion and overlap terms are *geometric regularizers*: they reduce ambiguous regions where an inlier for class y lies inside another class's ball, and they separate class prototypes in $d_{\mathbb{H}}$, which empirically helps open-set behavior when combined with s_{global}.
3) **Hyperbolic modeling choice.** The Poincaré ball embeds hierarchical or margin-heavy structure with bounded diameter; we claim it is a *useful* inductive bias for multi-class hypersphere packing, not that it is uniquely optimal for all tabular or image domains.
4) **Inference semantics.** With soft-boundary scores $s_k{=}d_{\mathbb{H}}^2{-}R_k^2$, minimizing s_k mimics choosing the "nearest" class ball; $\max(0, \min_k s_k)$ treats depth inside *all* balls as non-anomalous and flags points outside every ball—a consensus open-set heuristic, not a calibrated probability.

IV. Inference

Per-class score:

$$s_k(x) = d_{\mathbb{H}}(\mathbf{z}_k(x), \mathbf{c}_k)^2 - R_k^2. \tag{11}$$

Class prediction:

$$\hat{y}(x) = \arg\min_k s_k(x). \tag{12}$$

Global anomaly score:

$$s_{\text{global}}(x) = \max\left(0, \min_k s_k(x)\right). \tag{13}$$

If all s_k are positive, the sample lies outside every learned class sphere.

V. Implementation

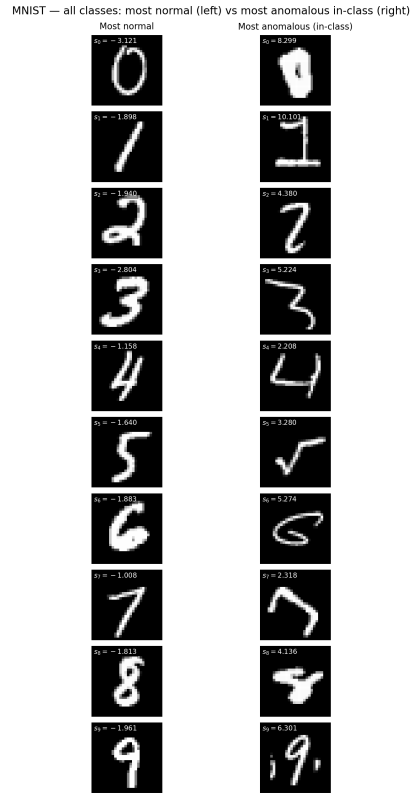
. follows Deep SVDD-style training (AE representation, fixed centers, no bias on last linear maps in the original SVDD reference) with extensions above. CHMS-SVDD implementation provides MNIST/CIFAR-10 training, evaluation, t-SNE, and Poincaré visualizations. Checkpoints store θ, $\mathbf{c}_k$, R_k, curvature c, and objective flags.

VI. Experiments (outline, requirements - Draft)

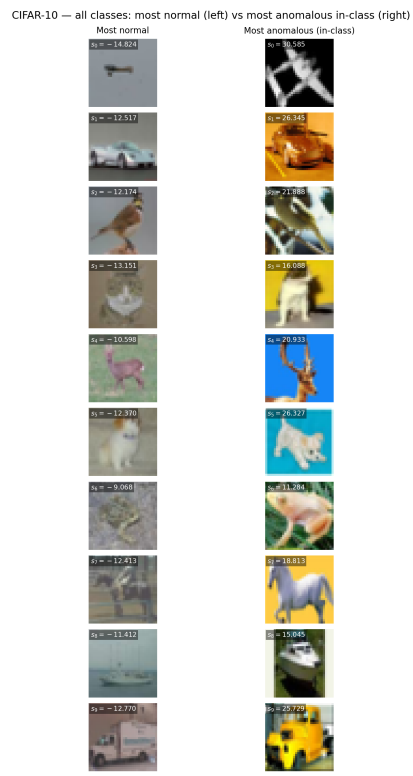
A. Datasets

MNIST and CIFAR-10 with $K{=}10$ and a one-class / open-set evaluation protocol.

Table **??** summarizes per-class one-vs-rest AUC for classical baselines and our hyperbolic-coupled variants (CHMS-SVDD and ablations).

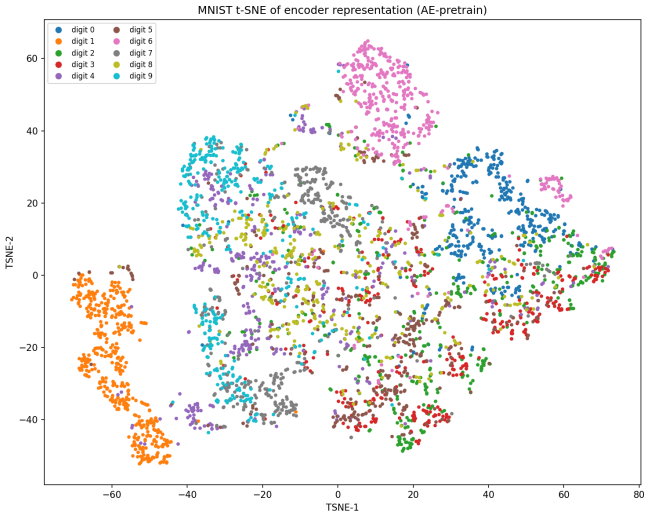


(a) MNIST: most normal (left) vs. most anomalous (right)

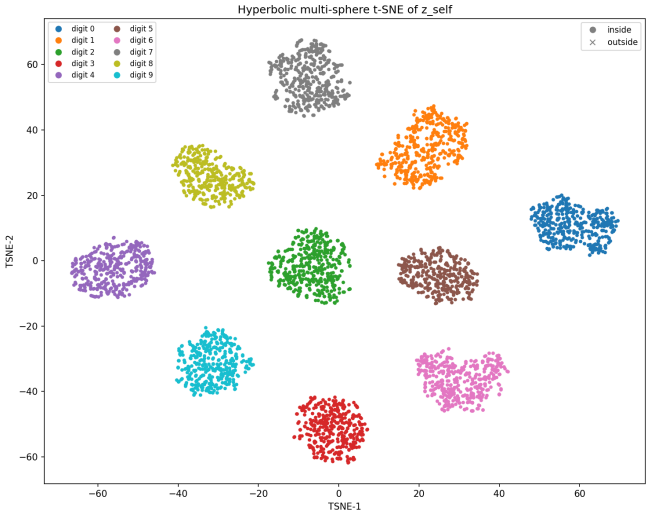


(b) CIFAR-10: most normal (left) vs. most anomalous (right)

Fig. 1: Most normal (left) and most anomalous (right) in-class test examples ranked by class-conditional soft-boundary scores $s_k = d_{\mathbb{H}}^2 - R_k^2$ (coupled hyperbolic multi-sphere model) for MNIST (top) and CIFAR-10 (bottom); see [1] for baseline Deep SVDD.



(a) AE-pretrain only: t-SNE of encoder representation (`rep`).



(b) After coupled SVDD: t-SNE of hyperbolic self-projection `z_self`.

Fig. 2: Why SVDD after AE pretraining is useful on MNIST. AE-pretrain (`rep`) shows partial class structure but substantial overlap, while coupled hyperbolic SVDD (`z_self`) yields tighter, class-consistent clusters with clearer separation.

AE pretraining learns a general representation, but class boundaries are still diffuse. The SVDD stage (with class-conditioned hyperbolic heads and sphere constraints) reshapes that space into compact class regions, which is exactly the structure used by the downstream anomaly score.

B. Metrics

Macro AUC from class-wise one-vs-rest scores; global anomaly ranking via s_{global}.

C. Ablations

Remove $\mathcal{L}_{\text{excl}}$ or $\mathcal{L}_{\text{ov}}$ and compare independent one-vs-rest models vs. the coupled model under hyperbolic scoring. The right block of Table **??** shows how CHMS-SVDD and its ablations trade off per-class AUC; on CIFAR-10, the `no_ov` variant often yields the highest class-wise AUC.

VII. Conclusion

We described CHMS-SVDD, a coupled hyperbolic multi-sphere extension of Deep SVDD that preserves class compactness while modeling inter-class exclusion and sphere separation, yielding a unified multiclass/open-set objective in a single model.

Acknowledgment

References

[1] L. Ruff, R. Vandermeulen, N. Goernitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, and M. Kloft, "Deep one-class classification," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 4393–4402.

[2] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2015.

TABLE II: Per-class anomaly detection AUC (%) for classical baselines and our hyperbolic-coupled variants. Left block: classical baselines. Right block: hyperbolic-coupled model (CHMS-SVDD) and ablations.

Dataset	**Class**	**OC-SVM**	**KDE**	**IF**	**DCAE**	**Deep SVDD**	**Hyperbolic-coupled**			
							CHMS-SVDD	**no_excl**	**no_ov**	**indep**
MNIST	0	98.6	97.1	98.0	96.6	98.0	**98.5**	98.4	98.5	98.3
	1	99.5	98.9	97.3	99.2	99.7	**99.9**	99.8	99.9	99.8
	2	82.5	79.0	88.6	85.0	91.7	**94.2**	88.4	89.2	83.1
	3	88.1	86.2	89.9	88.7	91.9	**94.2**	89.7	91.0	90.0
	4	94.9	87.9	92.7	89.4	94.9	95.3	93.3	**96.2**	92.7
	5	77.1	73.8	85.5	88.3	**88.5**	87.7	89.7	88.5	**91.2**
	6	96.5	87.6	95.6	94.7	98.3	**98.6**	97.9	97.8	96.8
	7	93.7	91.4	92.0	93.5	**94.6**	94.2	93.4	95.0	**95.5**
	8	88.9	79.2	89.9	84.9	**93.9**	**93.8**	91.3	92.6	91.4
	9	93.1	88.2	93.5	92.4	**96.5**	**95.5**	93.6	94.6	94.5
CIFAR-10	0 (Airplane)	61.6	61.2	60.1	67.1	61.7	78.1	63.8	**78.6**	62.2
	1 (Automobile)	63.8	64.0	50.8	54.7	65.9	69.4	44.3	**71.1**	44.4
	2 (Bird)	50.0	50.1	49.2	52.9	50.8	**73.0**	64.0	72.8	64.9
	3 (Cat)	55.9	56.4	55.1	54.5	59.1	66.3	59.0	**66.8**	58.9
	4 (Deer)	66.0	66.2	49.8	65.1	60.9	77.2	71.6	**77.9**	71.1
	5 (Dog)	62.4	62.4	58.5	60.3	65.7	**68.9**	62.8	67.4	62.6
	6 (Frog)	74.7	74.9	42.9	58.5	67.7	**80.8**	70.1	79.9	71.8
	7 (Horse)	62.6	62.6	55.1	62.5	67.3	71.5	58.5	**71.7**	57.3
	8 (Ship)	74.9	75.1	74.2	75.8	75.9	84.2	74.1	**84.3**	73.9
	9 (Truck)	75.9	76.0	58.9	66.5	**73.1**	71.6	50.3	**71.9**	48.8